



Predicting Flight Delays Using Machine Learning

DATASCI 261 FP Phase 3

Section 1 Group 4

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Meet the Team



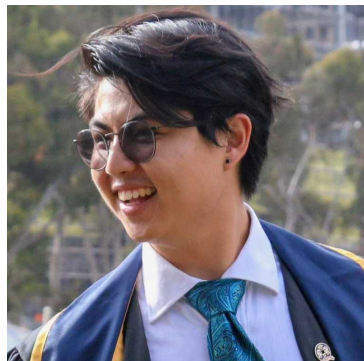
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Abstract

In a system this large and interconnected, effective delay prediction is necessary for mitigating disruption risks.



Scheduled passenger flights yearly:

9.8 Million



US public and private US airports:

19,482



Aircrafts in the sky at peak times:

5,500

Small delays can propagate through aviation networks, creating massive downstream problems.

- Delays disrupt aircraft routing, crew assignments, and gate operations, driving up operational costs for airlines.
- Extended delays (3+ hours) may trigger mandatory passenger compensation and service recovery costs.
- Flight delays degrade the passenger experience, threatening to brand loyalty and customer retention.

Scope of Phase 3

- 1 Project Description
- 2 Description of the Data
- 3 Exploratory Data Analysis (EDA)
- 4 Feature Engineering and Top Features
- 5 Overview of Modeling Pipelines Explored
- 6 Modeling Results
- 7 Conclusion

Project Description

To what extent can weather, congestion, and carrier operational patterns predict departure delays for US domestic flights?

The aim is to predict departure delays, so airlines can reduce disruptions and improve operational efficiency.

Input Variables

- CRS Elapsed Time, Distance
- Precipitation, Visibility, Wind Speed, Temperature
- Day of Week, Day of Month, Holiday Indicator
- Seconds Since Prev. Flight, Flights Last 7 Days, Airport Centrality

Output Variable

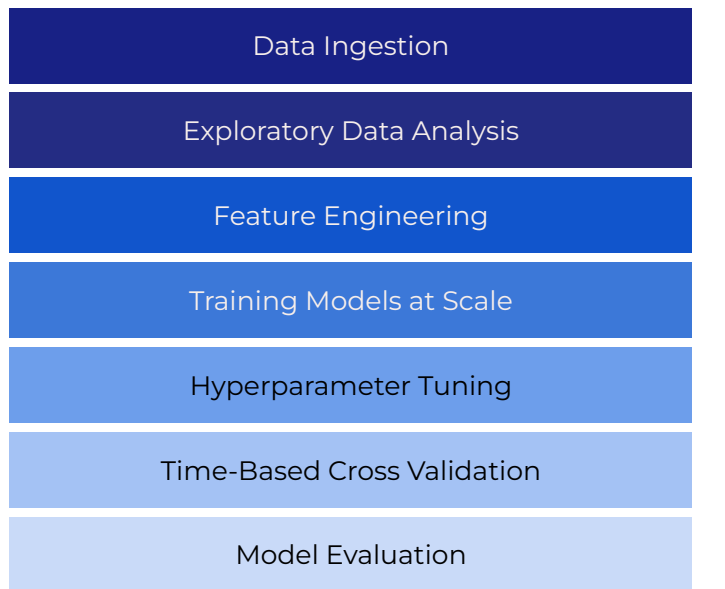
We divided delay times into four bins:

- Negligible delays, less than 15 minutes
- Moderate delays, 15-60 minutes
- Significant delays, 60-240 minutes
- Extreme delays, over 240 minutes

Models Used

- Baseline: Logistic Regression
- Multi-Layer Perceptron
- Random Forest
- Extreme Gradient Boosting

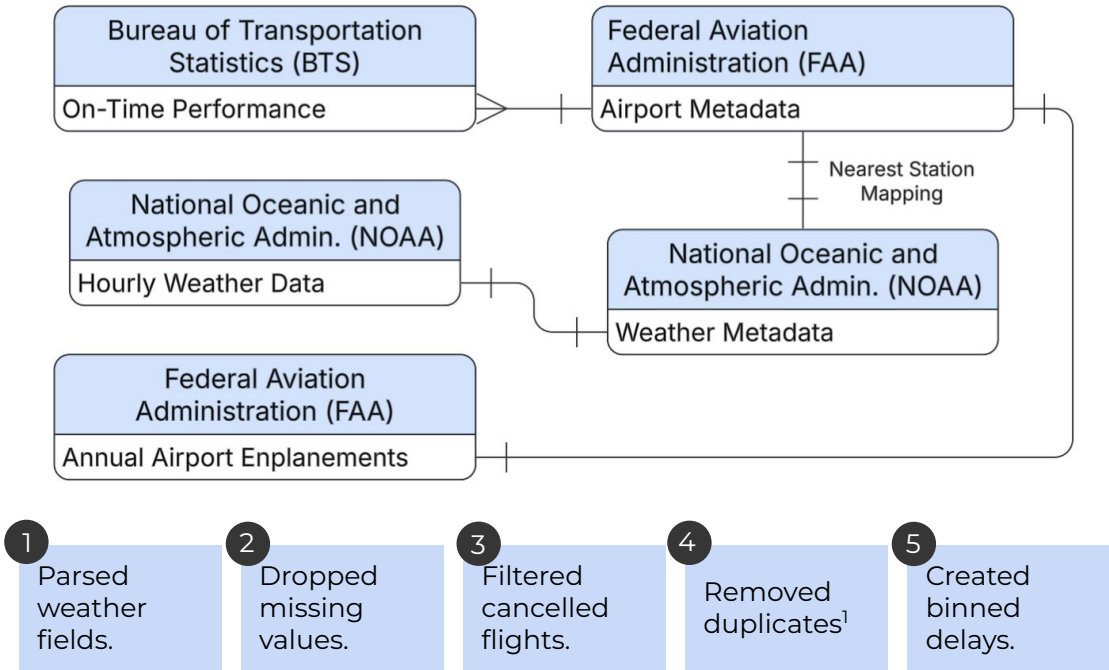
Our pipeline spans the full ML lifecycle, integrating 4 years of flight, weather, and airport traffic data.



Delay bins aligned to industry standards: delays <15 min. considered on-time; delays >240 min. may warrant passenger compensation.

Description of the Data

A custom integration pipeline consolidated flight records, hourly weather observations, and airport-level operational data into a unified modeling dataset.



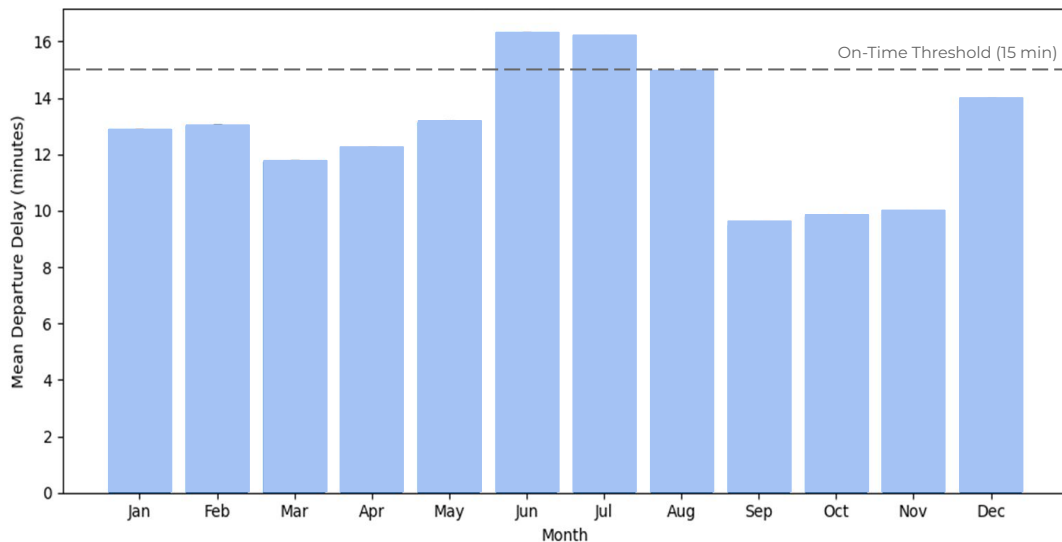
New features were engineered to reveal patterns not evident in the raw data.

- Seconds Since Previous Flight
- Number of Flights Last 7 Days
- Normalized Airport Centrality
- Nearest Holiday
- Holiday Window Code
- Days from Nearest Holiday

¹) Duplicates defined as any rows with the same flight date, departure time, & tail number.

Exploratory Data Analysis (EDA)

Delay severity exhibits clear seasonality, with trends mirroring travel demand.

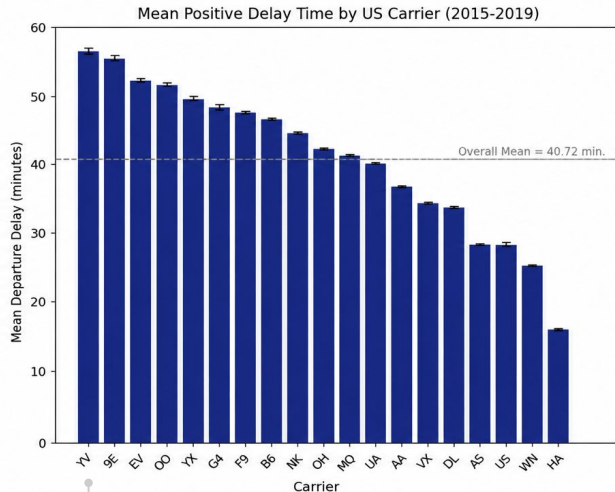


Key takeaways

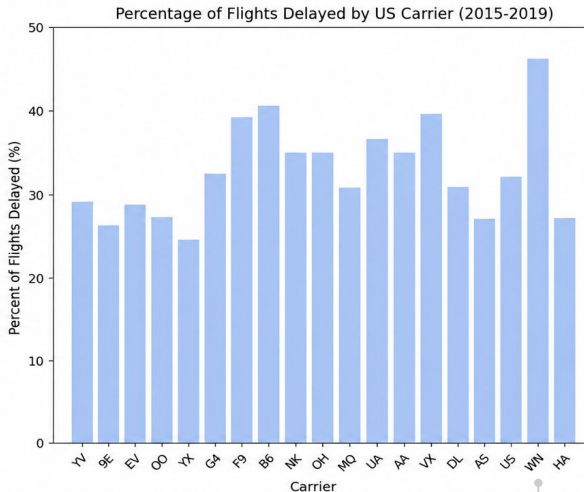
- Peak delays occur during popular travel seasons (summer and winter months), likely due to higher travel demand and operational congestion.
- The seasonal fluctuation justifies the use of month-based temporal features, which may capture meaningful patterns related to delay behavior.
- June and July are the only months with average delays exceeding the 15-minute BTS threshold.

Exploratory Data Analysis (EDA)

Delay frequency and severity do not exhibit a strong direct relationship across US carriers.



Mesa Airlines (YV) sees infrequent (29%) but severe (56 min.) delays, indicating that lower delay frequency does not necessarily translate to lower disruption severity.



Southwest Airlines (WN) has the 2nd lowest mean positive delay time (27 min.) but the highest percentage of flights delays (47%), likely due to their point-to-point model⁵.

Key takeaways

- Certain carriers (YV, 9E)¹ exhibit infrequent but severe delays, while others (WN, VX)² experience more frequent but shorter disruptions.
- Certain low-cost carriers (F9, B6)³ see both frequent and severe delays.
- The 'Big Three' airlines⁴ perform well overall, though consistent with rankings, Alaska Airlines achieves the strongest on-time performance.

1) Mesa Airlines, Endeavor Air

2) Southwest Airlines, Virgin America

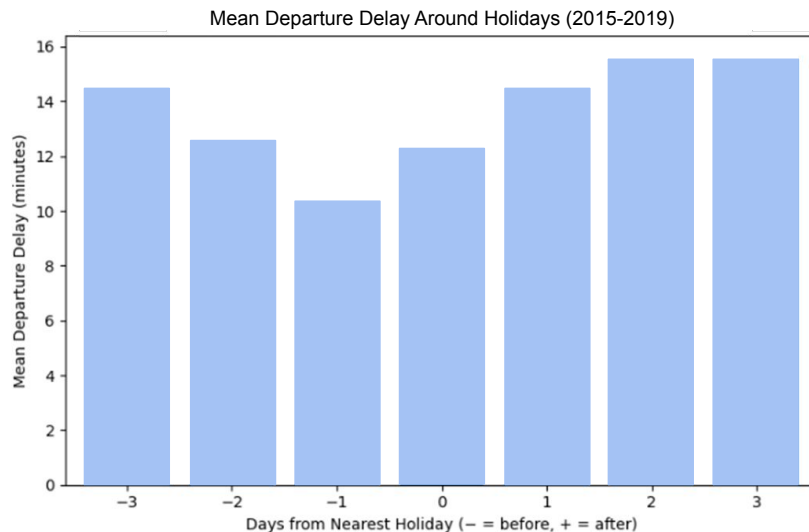
3) Frontier Airlines, JetBlue

4) Largest legacy carriers incl. American Airlines, Delta Airlines, and United Airlines.

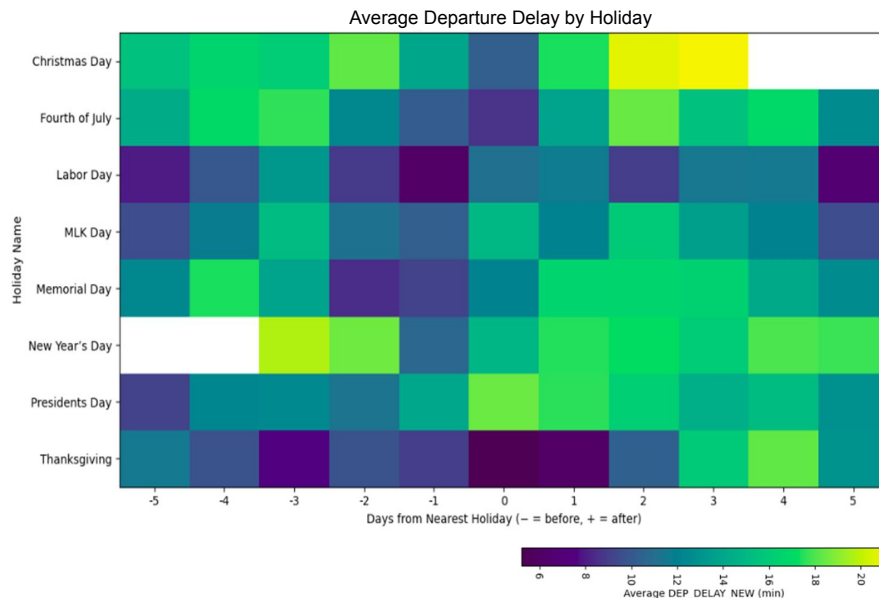
5) Direct routing model without centralized connection hubs

Exploratory Data Analysis (EDA)

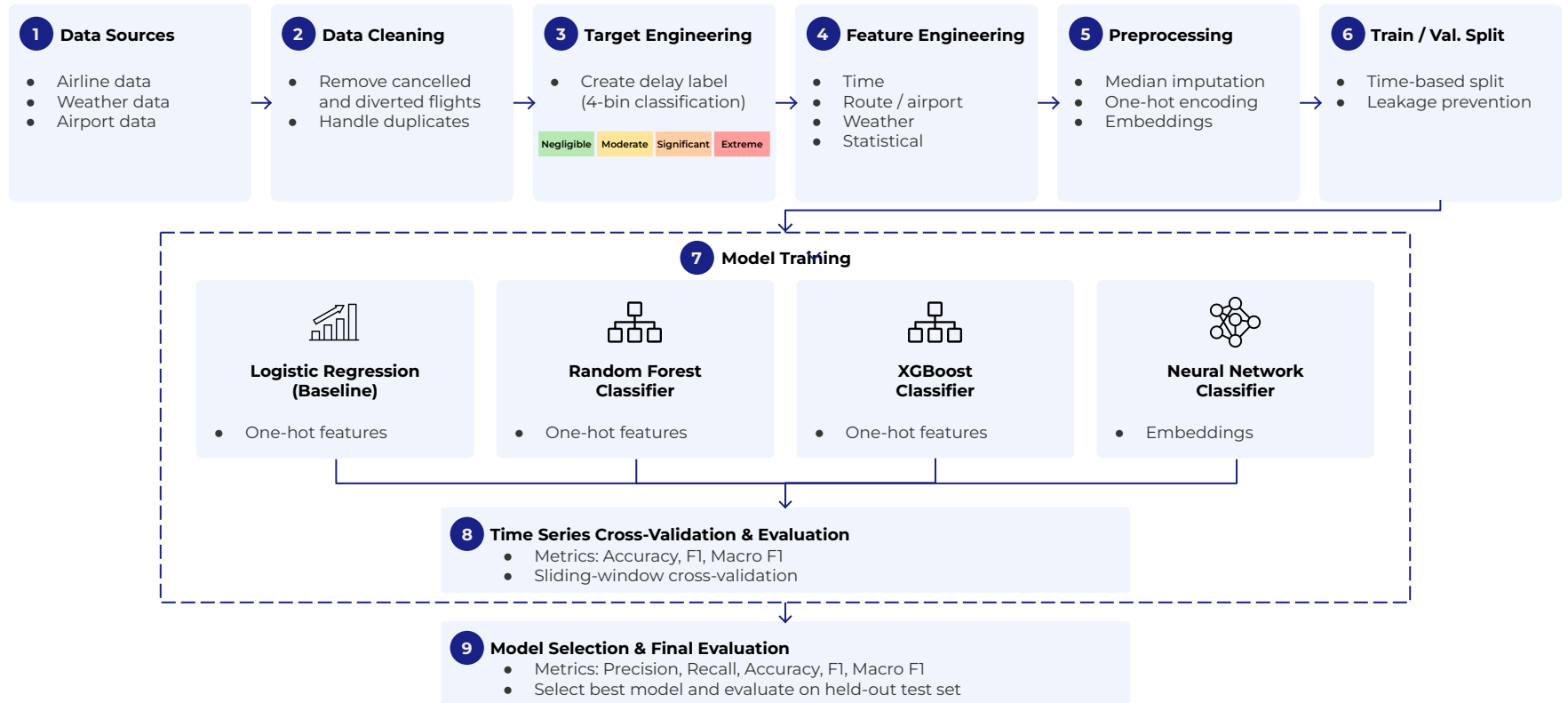
Delay times surrounding national holidays reflect pre- and post-holiday congestion as travelers move to and from destinations.



Holiday delay patterns align with seasonal travel behaviors and post-holiday return traffic surges.



Overview of Modeling Pipelines Explored

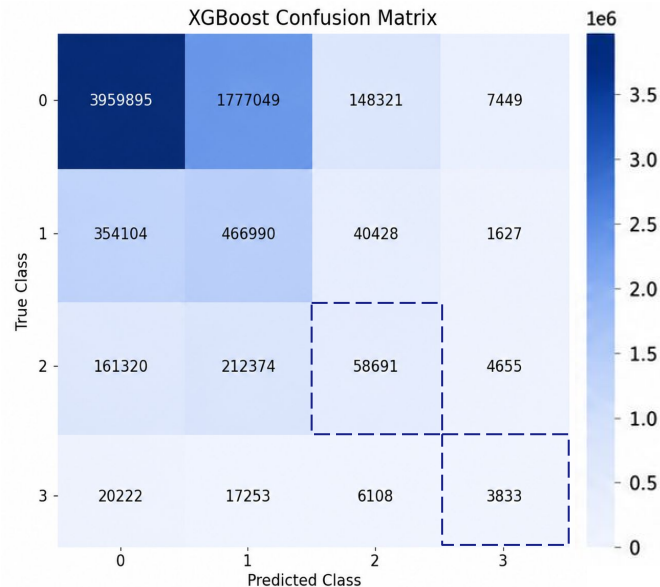
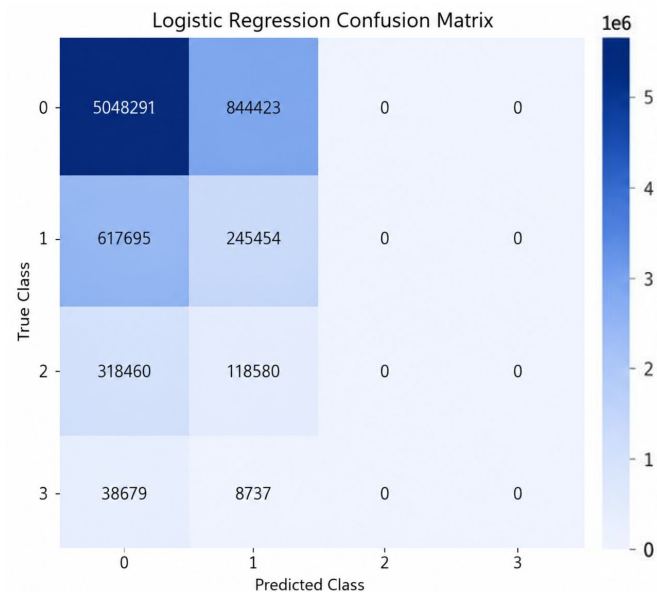


Modeling Results

Model	CV Accuracy	Training Accuracy	Testing Accuracy	CV Macro F1	Training Macro F1	Testing Macro F1
Naive Baseline	N/A	0.8200	0.8139	N/A	0.2253	0.2243
Logistic Regression	0.7333	0.7319	0.7311	0.2736	0.2753	0.2708
Multi-Layer Perceptron	0.8118	0.8189	0.8129	0.2304	0.2257	0.2248
Random Forest	0.7620	0.7639	0.7593	0.2803	0.2803	0.2754
XGBoost	0.6207	0.6222	0.6201	0.3258	0.3371	0.3326

Modeling Results

2019 Test Set



+62,524

additional
medium and
severe delay
cases captured
vs. baseline.

Driven by
improved
recognition of
minority classes.

Feature Importance

Rank	Feature	Category
1	Scheduled flight duration	 Flight details
2	Travel distance	
3	Hourly Precipitation	 Weather information
4	Hourly Visibility	
5	Hourly Wind Speed	
6	Hourly Dry Bulb Temperature	
7	Flight occurs on Friday	 Day-of-week
8	Flight occurs on Monday	

Conclusion

Key Takeaways

- **XGBoostClassifier:** Demonstrated the strongest ability to address class imbalance, leveraging features such as day of week and flight length, though with some tradeoff in overall accuracy.
- **Random Forest:** Bulb temperature and visibility were highly predictive, indicating that weather variables drive predictions.
- **Logistic Regression:** Origin airport was highly predictive, likely reflecting systematic delay patterns across specific airports.
- Class weighting and SMOTE did not improve weighted F1, as the model continued to favor the majority class..

Limitations & Future Work

- Class imbalance remained a significant modeling constraint, requiring stakeholders to determine the acceptable tradeoff between missed delays and false positive predictions.
- Incorporating real-time operational data (e.g., gate congestion) could improve responsiveness to dynamic delay conditions.

Thank you!
Questions?